Targeted transcript profiling of polyphenol oxidases in *Triticum* under environmental stress

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Introduction

Wheat is highly sensitive to abiotic stresses (drought, salinity, nutrient limitation), which impact both productivity and quality. These stresses triggers the accumulation of reactive oxygen species (ROS), leading to oxidative damage. Polyphenol oxidases (PPOs) are emerging as important in stress responses due to their role in ROS scavenging. Although traditionally associated with tissues browning, recent evidence challenges the assumption that low PPO activity is always desirable: modern cultivars with very low PPO levels often exhibit reduced pathogen resistance and greater susceptibility under combined stresses. In contrast, domesticated accessions and landraces with higher PPO activity tend to be more resilient. This suggests that overly suppressing PPOs may compromise plant resilience, highlighting the need for balanced PPO modulation to optimize stress tolerance and end-use quality. To investigate this hypothesis, we used targeted RNA sequencing (Ion AmpliSeq™) to profile PPO gene expression in *Triticum aestivum* L. and *Triticum turgidum* ssp. genotypes. Multiple tissues were analysed across four developmental stages under control and water-limited conditions (Figure 1), with and without plant growth–promoting bacterial consortia.

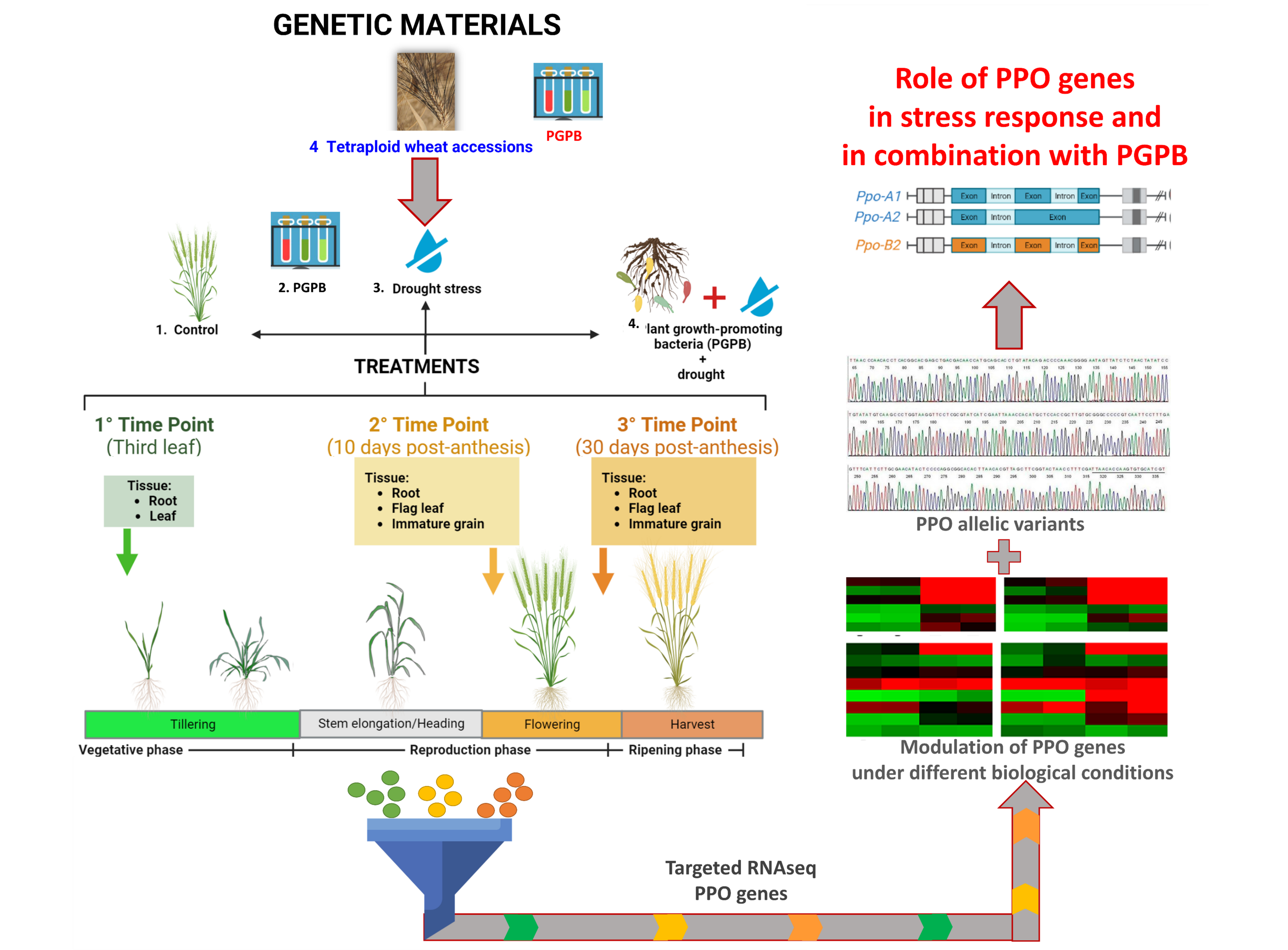

Figure 1. Experimental design for the modulation of PPO genes in wheat. A) Selection of wheat germplasm and morpho-agronomic evaluation under control, drought and drought + Plant Growth-Promoting Bacteria (PGPB); B) Collection of root, leaf and immature grain samples at three developmental stages, followed by RNA isolation; C) Targeted RNA sequencing of PPO genes; D) Construction of a PPO gene expression atlas and identification of allelic variants

Method

A custom Ion AmpliSeq™ panel (WG00631; Thermo Fisher Scientific) was developed to specifically amplify transcripts belonging to the PPO gene family in two wheat species: durum wheat (*Triticum turgidum* subsp. *durum* cv. Svevo) and bread wheat (*Triticum aestivum* cv. Chinese Spring). The panel was designed to target a total of 10 PPO transcripts, including 3 derived from *T. durum* and 7 from *T. aestivum*, using the reference genomic resources IWGSC RefSeq v2.1 for bread wheat and the Svevo genome assembly for durum wheat (Table 1). The WG00631 panel comprises 27 primer pairs designed for targeted amplification, with an average of three amplicons per PPO gene. Panel optimization was performed to enable reliable amplification and discrimination of closely related PPO transcripts.

Table 1. Genomic coordinates of target amplicons mapped onto the 10 PPO transcripts of Triticum turgidum and T. aestivum.

Transcript ID	Amp1	Amp2	Amp3
TRITD2Bv1G224170	392 - 620	710 - 939	1142 – 1316
TRITD2Av1G261300.2	416 - 644	734 - 963	1166 - 1343
TRITD2Av1G261390.1	422 - 650	-	1172 - 1346
TraesCS2A03G1101100	416 - 644	734 - 963	1166 - 1343
TraesCS2B02G41000	416 - 644	734 - 963	1166 - 1343
TraesCS2B03G1236300	416 - 644	734 - 963	1166 - 1343
TraesCS2D03G1045900	416 - 644	734 - 963	1166 - 1343
TraesCS2A03G1101800	422 - 650	-	1172 - 1346
TraesCS2D03G1047900	422 - 650	-	1172 - 1346
TraesCS2B03G1238100	437 - 665	755 - 984	1187 - 1361

RNA extracted from bread (Lankaoudali and Rebelde), durum wheat (Cappelli, Appulo, Svevo) and a *T. carthlicum* accession (PI115816) (three biological replicates per sample) was reverse-transcribed into cDNA and sequenced using Ion AmpliSeq™ technology. The resulting reads were aligned to the reference transcriptomes of Tri*cum turgidum* subsp. *durum* cv. Svevo and *Triticum aestivum* cv. Chinese Spring, following index generation with the TMAP (Torrent Mapping Alignment Program) suite. Reads mapped to the target regions were subsequently filtered using BED files containing the genomic coordinates of the targeted amplicons. Only reads with at least 90% overlap with the corresponding target region were retained for downstream analyses. Transcript abundance was estimated for each gene by selecting the amplicon with the highest read support, while redundant amplicons targeting the same transcript were excluded from quantitative analyses. Raw read counts were normalized using the TMM (Trimmed Mean of M values) method to account for compositional differences among libraries. Nucleotide variants were identified for each sample from the aligned sequencing data using FreeBayes and subsequently filtered, retaining only high-confidence calls with a read depth (DP) ≥ 10 and a variant quality score (QUAL) ≥ 30. The filtered variants were then annotated, and their potential effects on PPO structure and function were predicted using SnpEff.

Results and discussion

A bioinformatics workflow was developed to build a PPO gene expression atlas and identify allelic variants (Figure 2).

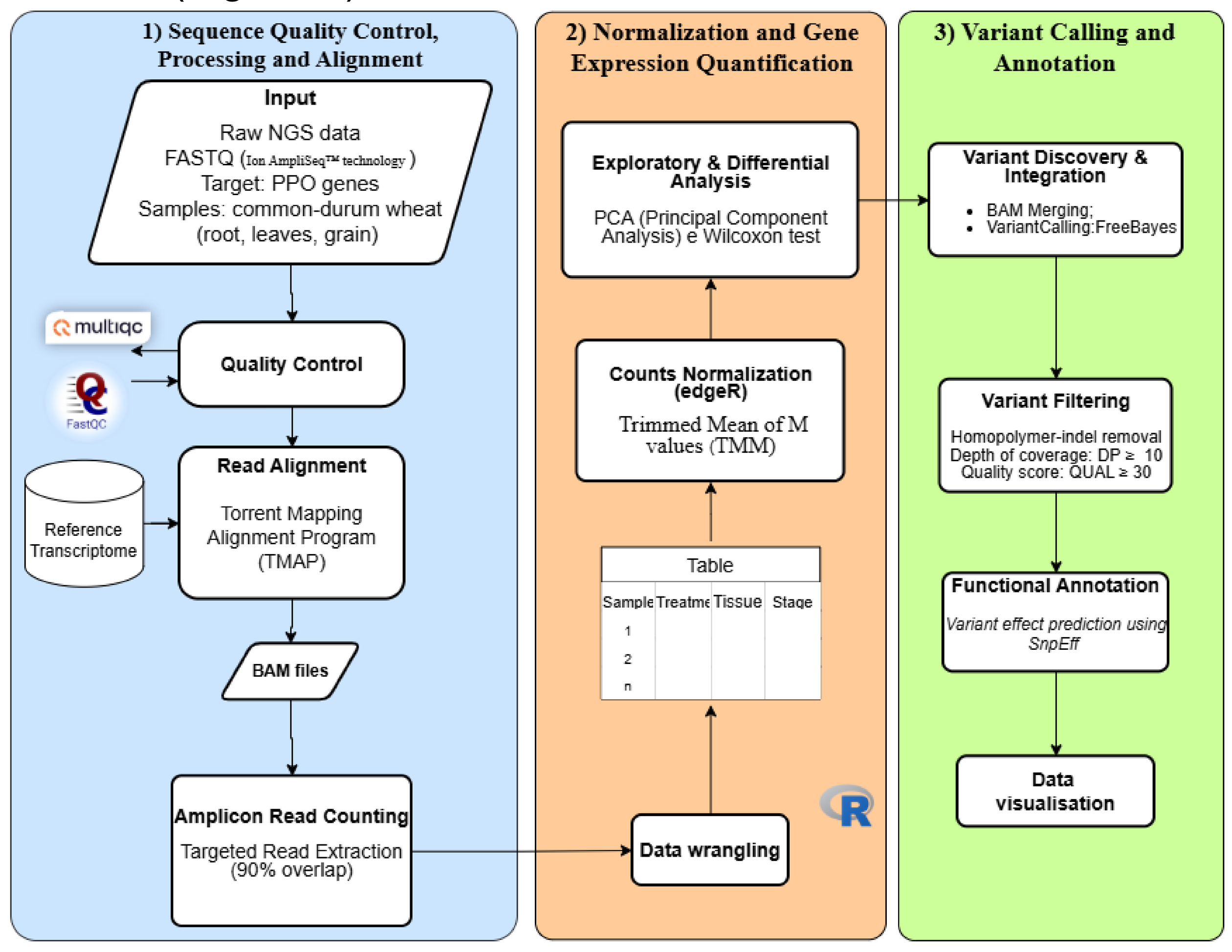

Figure 2. Workflow of the Ion AmpliSeq™ RNA-seq analysis. (1) Quality control and read alignment (FASTQ to BAM) using FastQC/MultiQC and TMAP, with targeted read extraction. (2) Gene expression analysis including data wrangling, TMM normalization (edgeR), and exploratory/differential analyses (PCA, Wilcoxon test). (3) Variant calling with FreeBayes, quality filtering (DP ≥ 10, QUAL ≥ 30), functional annotation with SnpEff, and data visualization.

Because raw amplicon counts are not directly comparable due to differences in amplification efficiency, they were treated as independent units, and gene expression was estimated using the best-performing amplicon per gene (Figure 3). Multiple amplicons mainly served to assess technical robustness of the assay. Substantial variability among biological replicates, partly driven by sequencing depth, highlighted the importance of proper normalization for reliable comparisons and robust downstream interpretation (Figure 4). Variants analysis enabled the identification of nucleotide polymorphisms within PPO loci (Figure 5) and the prediction of their potential effects on protein structure and function.

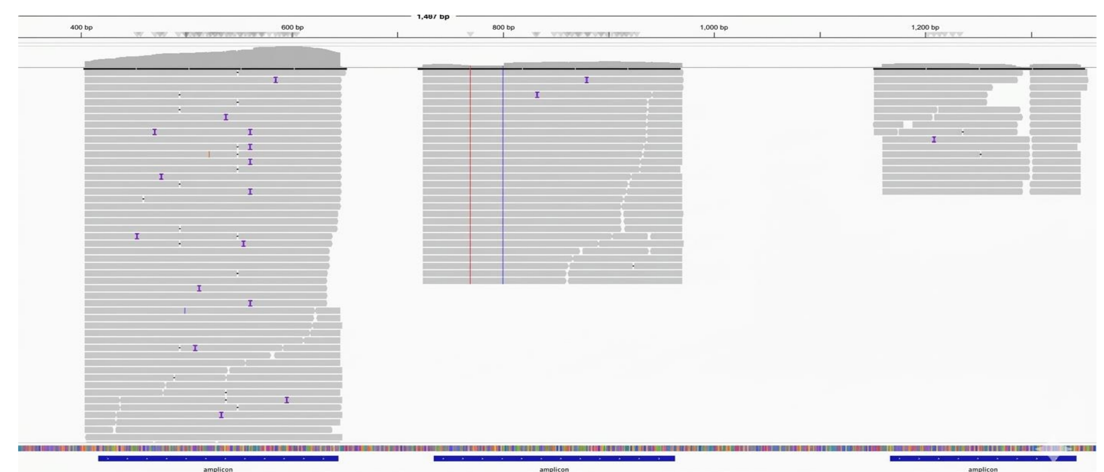

Figure 3. Visualisation of reads from targeted amplicons mapped to the TRITD2Bv1G224170.1 transcript.

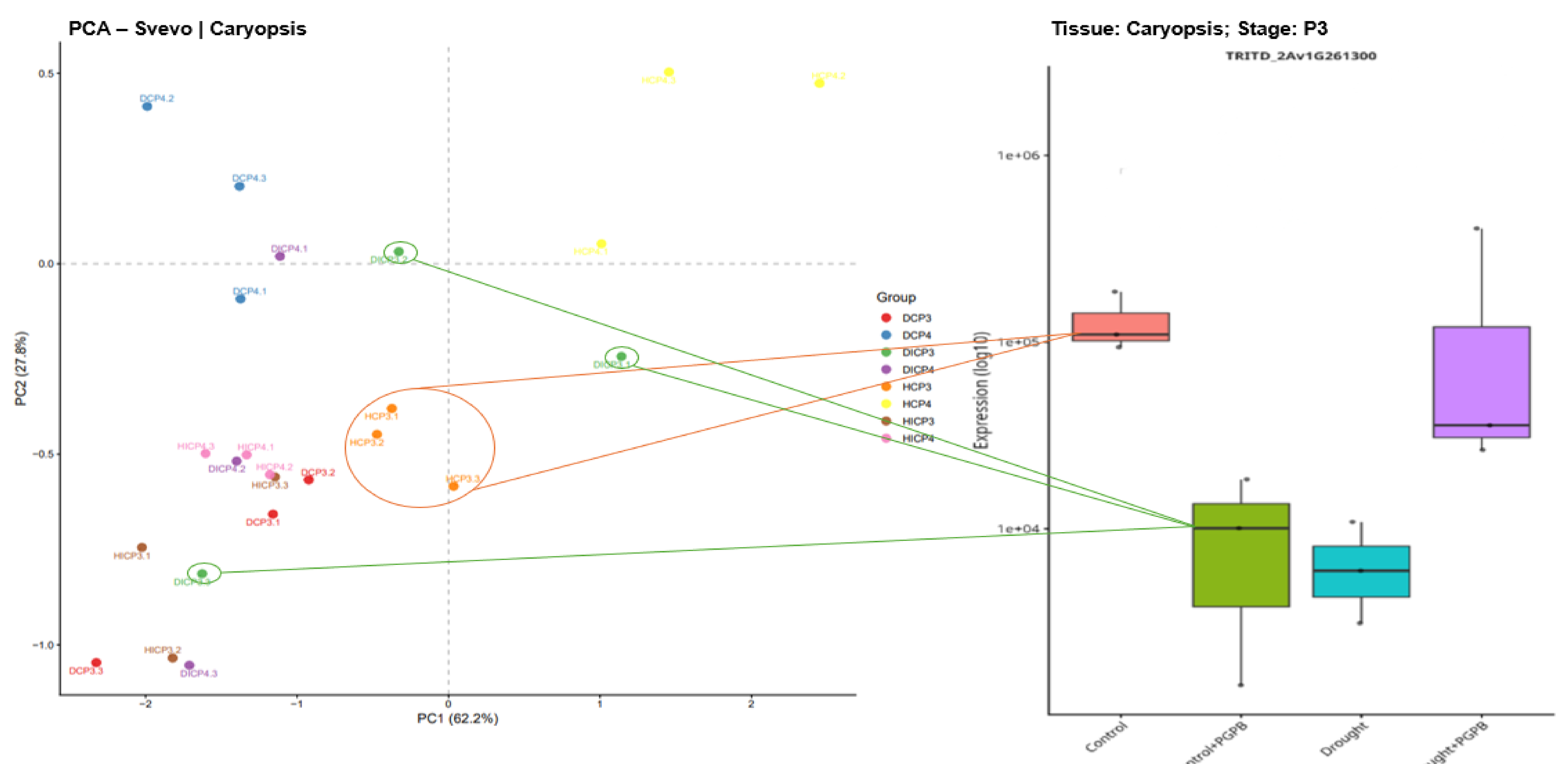

Figure 4. Principal component analysis (PCA) and distribution of gene expression levels in the Svevo variety (caryopsis tissue).

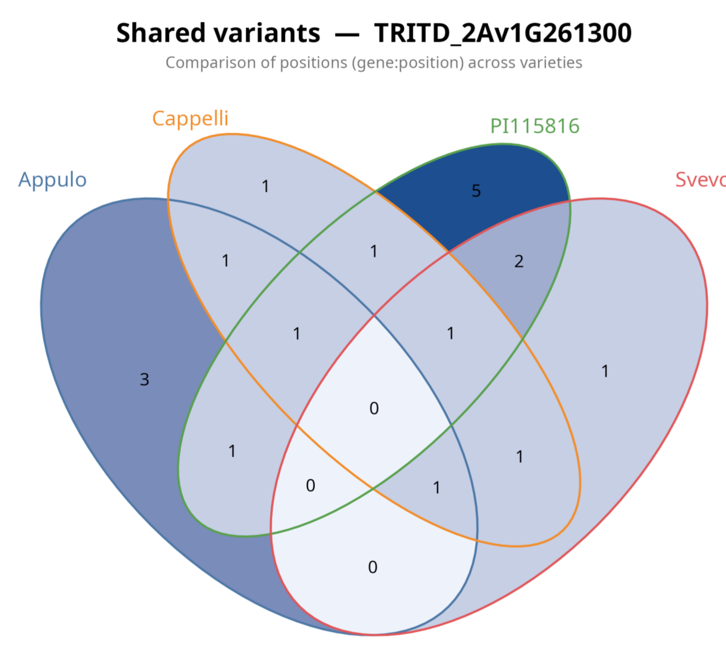

Figure 5. Venn diagram depicting the distribution of shared and unique variants identified in TRITD_2Av1G261300 across the Triticum turgidum spp. Appulo, Cappelli, PI115816, and Svevo.

Conclusion

Although Ion AmpliSeq™ was designed for diploid systems, we adapted it for polyploid wheat, where genomic redundancy poses challenges. Key issues included variability among replicates and primer efficiency, affecting gene expression estimates. To reduce bias, we used only the best-performing amplicon per gene. Despite these limitations, the approach proved robust for gene expression profiling and variant detection in complex wheat genomes.